

YatraVerse — Complete Project Brief

1. What Are We Building?

YatraVerse is an all-in-one tourism platform that connects **tourists, local guides, hotels, and restaurants** in one ecosystem.

Instead of tourists having to use different platforms for finding destinations, guides, hotels, restaurants, travel information, communication, and planning, YatraVerse brings these experiences together.

The platform also includes an **AI tourism assistant** and a **tourism points/leaderboard system** to make travel more personalized and engaging.

In one sentence:

YatraVerse connects travelers with the places, people, and services they need to experience a destination in one platform.

2. The Problem

Today, planning a trip often means jumping between different platforms.

A tourist might:

1. Search Google for a destination.
2. Look at Instagram or Facebook for local guides.
3. Search another platform for hotels.
4. Contact a guide through WhatsApp.
5. Contact a hotel through another platform.
6. Search separately for activities and attractions.
7. Manually create an itinerary.
8. Use an AI chatbot separately for travel questions.

This makes tourism **fragmented and time-consuming**.

There is also a problem on the business side.

Local guides, hotels, and restaurants may have good services but have difficulty getting discovered by tourists in one centralized tourism ecosystem.

Our problem statement

Tourism information, services, and communication are fragmented across different platforms, making travel planning difficult for tourists and making it harder for local tourism businesses to connect with travelers.

3. Our Solution

YatraVerse brings these different parts together.

A tourist can:

- Create an account.
- Explore destinations.
- Find local guides.
- View guide profiles.
- Hire/contact guides.
- Receive a custom schedule from a guide.
- Discover nearby hotels and restaurants.
- View prices, photos, ratings, and information.
- Chat with hotels.
- Send booking requests.
- Ask an AI tourism assistant questions.
- Earn points from exploring.
- Participate in a tourism leaderboard.
- Review guides and businesses.

At the same time:

Guides can:

- Create an account.
- Create a professional profile.
- Specify their location.
- Specify areas they cover.
- Explain their experience and services.
- Set availability.
- Connect with tourists.
- Create customized schedules for tourists.
- Receive reviews and ratings.

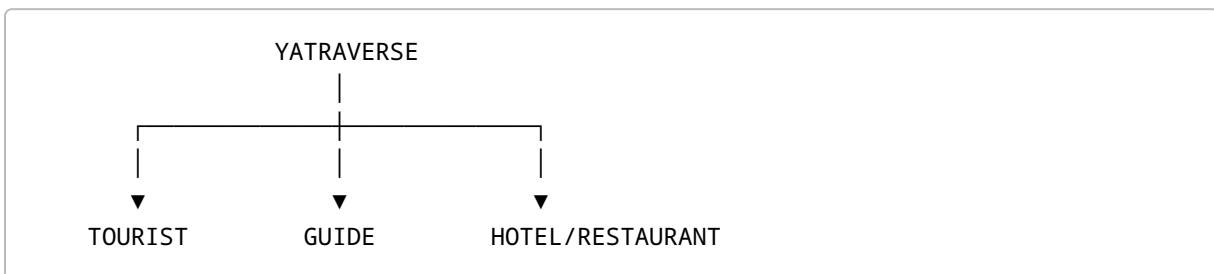
Hotels/Restaurants can:

- Register their business.
- Add their location.
- Upload photos.

- Add descriptions.
 - Add prices.
 - Add available services.
 - Display ratings.
 - Communicate with tourists.
 - Receive booking requests.
-

4. The Three Main User Types

Our system has three primary user roles.



Each role gets a different dashboard and different capabilities.

5. Tourist Experience

Step 1 — Create Account

A tourist registers on YatraVerse.

They can have:

- Name
- Profile picture
- Email
- Password
- Interests
- Travel preferences

After logging in, they can access the tourism platform.

6. Tourist Dashboard

The tourist dashboard can show:

Welcome, Prail

Explore Nepal

[Kathmandu]

[Pokhara]

[Chitwan]

[Mustang]

[Lumbini]

Your Points: 1,250

Your Rank: #17

Upcoming Trip

Pokhara – 3 Days

AI Assistant

The dashboard is essentially the tourist's control center.

7. Destination Exploration

The tourist selects a destination.

For example:

POKHARA

The destination page can show:

- Description
- Attractions
- Activities
- Photos
- Location
- Available guides
- Nearby hotels
- Nearby restaurants

- Ratings
- Recommended experiences

The important idea is that **selecting a destination connects the tourist to the services available there.**

8. Finding a Local Guide

Suppose the tourist selects Pokhara.

YatraVerse finds guides registered around Pokhara.

The tourist can see:

LOCAL GUIDES

Guide A

★★★★★

Adventure + Hiking

Pokhara

Available

[\[View Profile\]](#)

Guide B

★★★★☆

Culture + Photography

Pokhara

Available

[\[View Profile\]](#)

The tourist can open a guide's profile.

9. Guide Profile

The guide profile can contain:

GUIDE PROFILE

Name

Profile Photo

About Me

Location:
Pokhara

Experience:
5 years

Specialties:

- Hiking
- Local Culture
- Photography

Languages:
Nepali
English

Areas Covered:
Pokhara
Sarangkot
Begnas

Rating:
4.8 / 5

[Contact]
[Hire Guide]

The goal is to help tourists understand **who the guide is and what they can provide** before hiring them.

10. Custom Guide Schedule

One of the important features is personalized scheduling.

A guide can create a schedule for a tourist.

For example:

POKHARA — 3 DAY EXPERIENCE

DAY 1

08:00 — Hotel pickup

09:00 – Phewa Lake
11:00 – World Peace Pagoda
13:00 – Lunch
15:00 – Lakeside exploration

DAY 2

05:00 – Sarangkot sunrise
08:00 – Breakfast
10:00 – Adventure activity
14:00 – Local experience

DAY 3

08:00 – Begnas Lake
11:00 – Local village
15:00 – Return

The tourist can see this schedule and understand exactly what the guide is proposing.

11. Hotel and Restaurant Discovery

When a tourist selects a destination, YatraVerse can also show nearby hotels and restaurants.

For example:

POKHARA

Hotels

Hotel A

★★★★★

NPR XXXX/night

Hotel B

★★★★☆

NPR XXXX/night

Hotel C

★★★★☆

NPR XXXX/night

Each business profile can contain:

- Photos
- Location
- Description
- Price
- Facilities
- Services
- Rating
- Reviews
- Contact
- Booking option

12. Hotel Profile

Example:

HOTEL PROFILE

Hotel Himalayan

★★★★★ 4.7

Pokhara, Nepal

About:

...

Facilities:

- ✓ WiFi
- ✓ Breakfast
- ✓ Parking
- ✓ Restaurant

Starting from:

NPR XXXX/night

[Chat]

[Request Booking]

The hotel can manage its own information through its dashboard.

13. Tourist ↔ Hotel Chat

A tourist can communicate directly with a hotel.

Example:

Tourist:

Do you have a room available
for two people from August 12
to August 15?

Hotel:

Yes, we have a deluxe room
available for those dates.

The idea is to keep communication inside the platform.

The same concept can eventually be extended to:

Tourist ↔ Guide
Tourist ↔ Hotel

14. Booking

The tourist can send a booking request.

Example flow:

Tourist
↓
Select Hotel
↓
View Hotel
↓
Select Dates
↓
Request Booking

```
↓  
Hotel receives request  
↓  
Hotel accepts/rejects  
↓  
Tourist receives status
```

Important for the 24-hour hackathon:

We should implement **booking requests**, not a complicated payment system.

We don't need to build:

- Payment gateway
- Refund system
- Complex cancellation system
- Financial transaction processing

Those can be future features.

15. Guide Registration

A guide can register as a different account type.

During registration:

Account Type:

- Tourist
- Guide
- Hotel / Restaurant

If they select Guide, they provide information relevant to their profession.

The guide dashboard allows them to:

- Edit profile
- Add location
- Add areas covered
- Add services
- Add experience
- Set availability
- View tourist requests

- Create schedules
 - Manage their profile
-

16. Hotel/Restaurant Registration

A business can register similarly.

They provide:

- Business name
- Business type
- Location
- Description
- Photos
- Services
- Prices
- Contact information

The business then receives its own dashboard.

17. AI Tourism Assistant

YatraVerse also contains an AI chatbot powered by the **Mistral API**.

The tourist can ask:

What should I visit in Pokhara?

I have three days in Kathmandu. What should I do?

I like adventure and nature. Which destination should I choose?

What activities are available in Pokhara?

Help me plan my trip.

The chatbot acts as a **tourism assistant**, not simply a generic chatbot.

Eventually, it can use the platform's destination, guide, and hotel data to give more contextual recommendations.

18. Tourism Points System

We also want to make tourism more engaging.

Tourists can earn points from legitimate tourism activities.

For example:

Explore new destination	+100
Complete a trip	+150
Complete guide experience	+75
Verified review	+25
Discover new attraction	+50

The exact scoring system can be adjusted.

19. Leaderboard

The points create a tourism leaderboard.

TOURISM LEADERBOARD

#1	Tourist A	4,850 pts
#2	Tourist B	4,420 pts
#3	Tourist C	4,100 pts
#4	Tourist D	3,870 pts

This encourages users to explore more places and participate in different tourism experiences.

Important:

We should avoid allowing users to simply click a button repeatedly to earn points.

Points should eventually be tied to legitimate/verified actions.

For the MVP, we can keep the system simple.

20. Reviews and Ratings

After interacting with a guide or hotel, tourists can leave:

- Rating
- Review

Example:

★★★★★ 4.8

"Very knowledgeable guide and excellent local recommendations."

These ratings help future tourists decide who to trust.

They also create a reputation system for guides and businesses.

21. Database

The database is extremely important because almost everything in YatraVerse is connected.

We are using:

Supabase + PostgreSQL

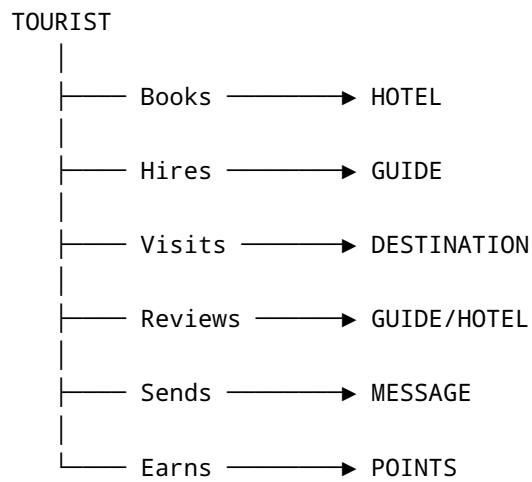
The database will store things like:

Users
Tourists
Guides
Hotels
Restaurants
Destinations
Trips
Schedules
Bookings
Messages
Reviews
Ratings

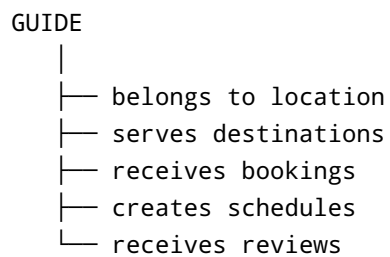
Points
Leaderboard data

22. How the Database Connects Everything

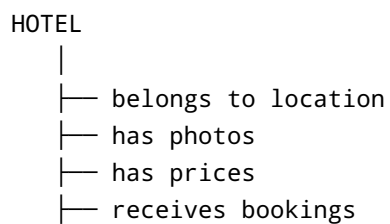
For example:



And:



And:



```
└─ chats with tourists
└─ receives reviews
```

This is why PostgreSQL is appropriate for the project.

23. Backend

We will use:

Node.js + Express.js

The backend sits between the frontend and our services/database.

```
React
  ↓
Node.js + Express
  ↓
Supabase
```

For example, when a tourist wants to see guides in Pokhara:

```
React
  ↓
GET /api/guides?location=pokhara
  ↓
Node.js
  ↓
Supabase
  ↓
Guides returned
  ↓
React displays them
```

24. AI Backend

The Mistral API key should **not** be placed directly in the React frontend.

Correct architecture:

```
React
  ↓
Node.js / Express
  ↓
Mistral API
  ↓
Response
  ↓
React
```

The API key stays in backend environment variables.

25. Frontend Technology

We will use:

- React.js
- Vite
- Tailwind CSS

React handles the application.

Tailwind handles the styling.

Vite handles the development/build environment.

We are currently **not using the 3D globe**, so we don't need Three.js, React Three Fiber, GSAP, or GeoJSON for the MVP.

26. Complete Tech Stack

Frontend

```
React.js
Vite
Tailwind CSS
JavaScript / JSX
```


Backend

Node.js
Express.js
REST API
JSON

Database

Supabase
PostgreSQL

Authentication

Supabase Auth

Storage

Supabase Storage

AI

Mistral API

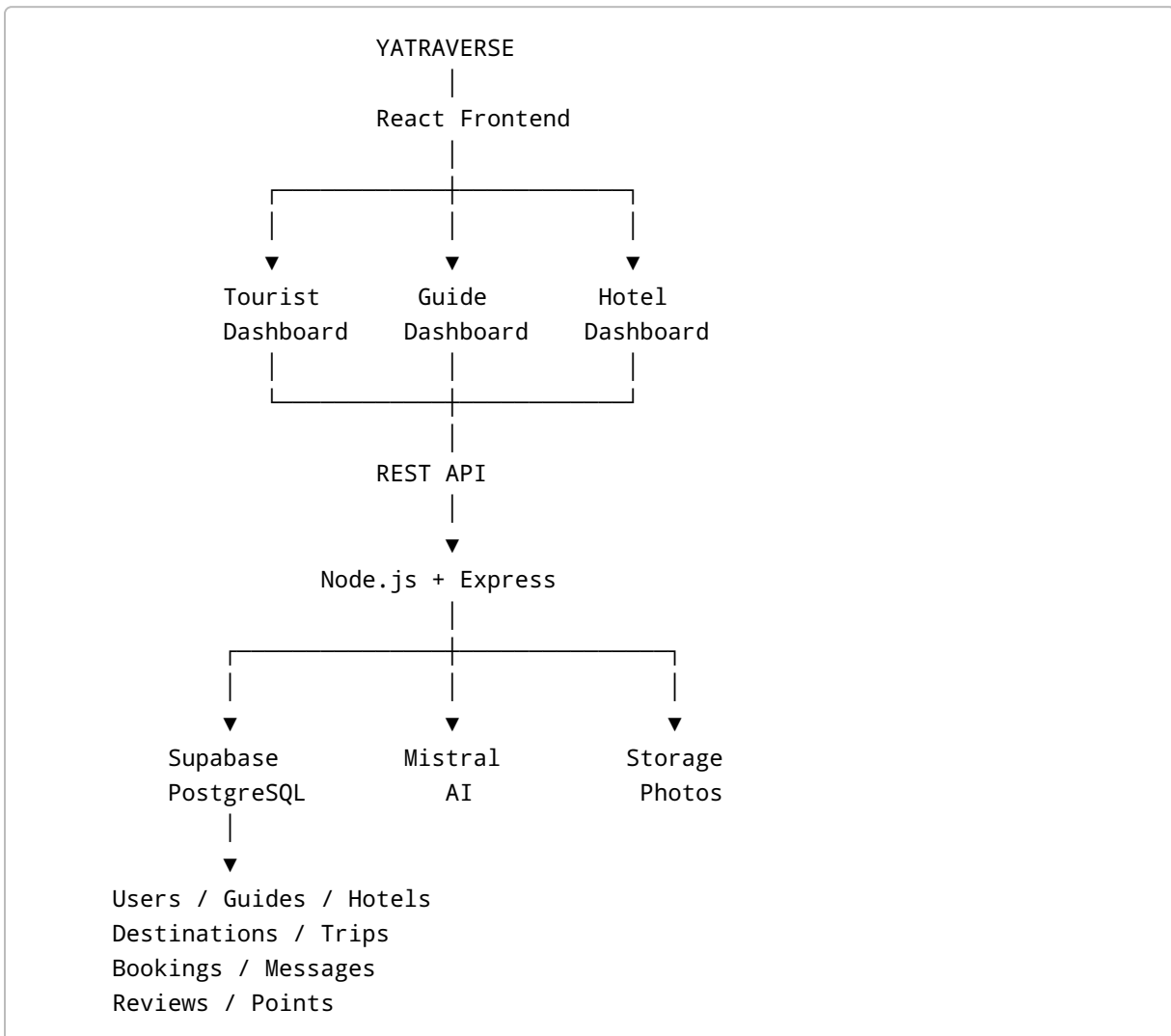
Development

Git
GitHub
Antigravity

Deployment

Vercel
Node.js hosting
Supabase

27. Complete Architecture



28. 24-Hour MVP

We have a large vision, but we cannot build everything perfectly in 24 hours.

Our priority is to create a **working demonstration of the core ecosystem**.

MUST HAVE

Tourist

- Registration/login

- Explore destinations
- View guides
- View hotels
- View profiles
- Send basic booking/contact requests

Guide

- Registration
- Profile
- Location
- Services
- Basic schedule creation

Hotel

- Registration
- Business profile
- Location
- Photos
- Price
- Booking request

Platform

- AI chatbot
- Basic points system
- Leaderboard
- Basic ratings/reviews
- Responsive UI

29. Features We Can Add If Time Remains

If the core system is working:

- Tourist ↔ Hotel real-time chat
 - Tourist ↔ Guide chat
 - Advanced filtering
 - Better itinerary builder
 - Hotel availability
 - More advanced recommendations
 - Better leaderboard
 - Notifications
 - More detailed reviews
-

30. Features for Future Versions

We should NOT spend our 24 hours trying to build everything.

Future versions can include:

- Online payments
 - Live location tracking
 - Maps/navigation
 - Verified guide identity
 - Fraud detection
 - Advanced AI recommendations
 - Multi-language support
 - Push notifications
 - Business analytics
 - Government tourism integration
 - Complete booking management
 - Cancellation/refund systems
-

31. What We Need to Demonstrate to Judges

The most important thing is that the judges can understand the ecosystem quickly.

Our demo should show:

1. Tourist creates account
- ↓
2. Selects a destination
- ↓
3. Finds a local guide
- ↓
4. Views guide profile
- ↓
5. Finds nearby hotel
- ↓
6. Views hotel profile
- ↓
7. Sends booking request
- ↓
8. Uses AI tourism assistant
- ↓
9. Gets points

↓
10. Appears on leaderboard

At the same time, we can briefly demonstrate:

Guide → creates profile → creates schedule

Hotel → creates profile → receives booking request

That proves that this isn't just a tourism information website.

It is an **ecosystem connecting different participants.**

32. What Makes YatraVerse Different?

Our main differentiator is the combination of:

Tourist + Guide + Hotel + Restaurant + Destination + AI + Communication + Booking + Gamification

Most individual platforms focus heavily on one part of tourism.

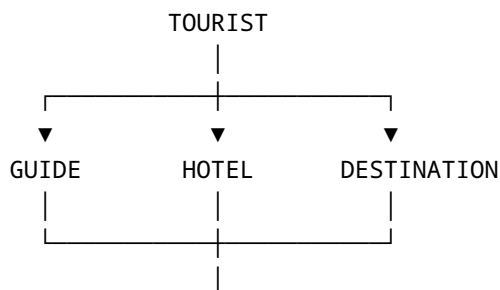
YatraVerse attempts to connect the different participants in one ecosystem.

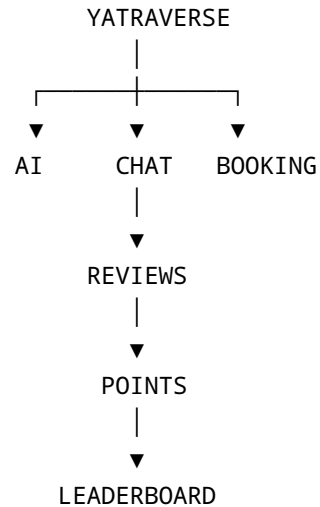
The core idea is:

Don't make the tourist search for the tourism ecosystem. Bring the tourism ecosystem to the tourist.

33. Team Mental Model

Everyone on the team should understand this simple structure:





If every team member understands this flow, they can work on different parts without losing sight of the overall product.

34. The One-Line Pitch

YatraVerse is an all-in-one tourism ecosystem that connects travelers with local guides, hotels, restaurants, destinations, and AI-powered travel assistance in one platform.

35. The Core Message

We are not trying to build **just another travel website**.

We are trying to make the tourism experience **connected**.

A tourist should be able to enter one platform and go from:

"Where should I go?"

to

"Who can guide me?"

to

"Where can I stay?"

to

"Can I book it?"

to

"What should I do there?"

to

"How was my experience?"

without having to leave the ecosystem.